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Experiment No-01

Study of CMOS NOT Gate (Inverter) Schematics in Cadence Virtuoso Software

1.1 Objectives

The main objectives of this experiment are summarized as:

- To understand the concept and operation of a CMOS NOT gate (inverter) as the fundamental building block of digital logic circuits.
- To become familiar with the Cadence Virtuoso schematic editor and the standard IC design flow used to create standard cells.
- To design the transistor-level schematic of a CMOS NOT gate using NMOS and PMOS devices from the gpd90 technology library, and to generate its schematic symbol.
- To construct a test bench for the NOT gate and verify its functionality through DC and transient simulation using Spectre in the Analog Design Environment (ADE L).
- To obtain and analyze the voltage transfer characteristic (VTC) and the transient input/output wave-forms of the designed NOT gate.

Keywords- CMOS NOT gate, NMOS, PMOS, Cadence Virtuoso, schematic editor, Spectre, voltage transfer characteristic (VTC), truth table, standard cell.

1.2 Theory

A NOT gate, also called an inverter, is the simplest logic circuit in digital electronics. It has a single input and produces an output that is the logical complement of that input, expressed by the Boolean expression $Y = A'$. Whenever the input is LOW (logic 0), the output is HIGH (logic 1), and whenever the input is HIGH (logic 1), the output is LOW (logic 0). The truth table of an ideal NOT gate is shown in Table-1.1.

A (Input)	$Y = A'$ (Output)
0	1
1	0

Table-1.1. Truth table of an ideal logic NOT gate (inverter).

1.2.1 The CMOS NOT Gate (Inverter)

A Complementary Metal-Oxide-Semiconductor (CMOS) inverter implements the NOT function using a complementary pair of transistors: a PMOS transistor connected between the output node and the supply voltage VDD (the pull-up network), and an NMOS transistor connected between the output node and ground, VSS (the pull-down network). The gates of the two transistors are tied together to form the single input terminal, and their drains are tied together to form the output terminal.

When the input is LOW (logic 0), the NMOS transistor is OFF and the PMOS transistor is ON, so the output node is pulled up to VDD, producing a HIGH output. When the input is HIGH (logic 1), the NMOS transistor turns ON and the PMOS transistor turns OFF, pulling the output down to ground and producing a LOW output. Because exactly one of the two transistors is ON in either steady state and the other is OFF, there is no direct conduction path from VDD to

ground, and the static power dissipation of a CMOS gate is ideally zero; power is dissipated mainly during switching (dynamic power) and due to leakage currents in real devices.

1.2.2 Voltage Transfer Characteristic (VTC) and Transistor Sizing

The Voltage Transfer Characteristic (VTC) of an inverter is a plot of the output voltage (V_{out}) as a function of the input voltage (V_{in}), obtained from a DC sweep of the input. The point on this curve where V_{in} equals V_{out} is called the switching threshold voltage, V_M . For a symmetric and noise-robust design, V_M is usually targeted close to $V_{DD}/2$, and the transition from HIGH to LOW should be as sharp as possible.

Because the mobility of holes in a PMOS transistor is lower than the mobility of electrons in an NMOS transistor, a PMOS device must be made wider than an NMOS device of the same length to provide comparable pull-up and pull-down drive strength. A commonly used sizing rule, also followed in this experiment, is to make the PMOS width approximately two to three times the NMOS width (a ratio of 3:1 is used here, with $W_P = 720$ nm and $W_N = 240$ nm for the same channel length of 100 nm). Proper sizing balances the rise time and fall time of the output and keeps the switching threshold close to the midpoint of the supply range.

1.2.3 Cadence Virtuoso Design Flow

In this experiment, the CMOS NOT gate is designed and verified using the Cadence Virtuoso IC design environment. The overall flow consists of creating a design library attached to a process technology library (gpd090), drawing the transistor-level schematic in the Schematic XL editor by placing and sizing NMOS and PMOS instances and wiring them together, generating a schematic symbol for the NOT gate cell, and finally building a separate test bench schematic in which the NOT gate symbol is instantiated along with input and supply sources. The test bench is then simulated using the Spectre simulator inside the Analog Design Environment (ADE L) to verify the NOT gate's transient and DC behaviour.

1.3 Hardware/Software Used for the Experiment

1. Cadence Virtuoso Schematic Editor (Schematic XL)
2. Virtuoso Analog Design Environment (ADE L)
3. Spectre circuit simulator
4. gpd090 generic process design kit (technology library)
5. NMOS and PMOS transistors (gpd090_nmos1v, gpd090_pmos1v)
6. Pulse voltage source (vpulse), DC voltage source (vdc) and ground (gnd) from analogLib
7. Linux workstation running the Cadence design environment

1.4 Experimental and Simulation Analysis

A new design library (NOT_gate) was created and attached to the gpd090 technology library, and a schematic cellview named NOT Inverter was created within it. NMOS and PMOS

instances were placed in the schematic, sized as described in Section 1.2.2, and wired together with input, output and supply pins. A schematic symbol was then generated for the NOT gate cell. A separate test schematic (NOT Inverter_test) was built in which the NOT gate symbol was instantiated with a pulse source at its input and a DC supply and ground connected to its power pins, and the design was simulated in ADE L using the Spectre simulator. Each of these steps is illustrated below.

1.4.1 Transistor Sizing and Parameter Selection

The width and length of each transistor were set at the time of placement in the Virtuoso Schematic XL editor. The NMOS transistor was sized with a total width of 240 nm and a channel length of 100 nm, while the PMOS transistor was sized with a total width of 720 nm and the same channel length of 100 nm, giving the 3:1 PMOS-to-NMOS width ratio discussed in Section 1.2.2. The complete set of device parameters used in the design is summarised in Table-1.2.

Device	Width (W)	Length (L)
NMOS (gpdk090_nmos1v)	240 nm	100 nm
PMOS (gpdk090_pmos1v)	720 nm	100 nm

Table-1.2. Device parameters used for the NMOS and PMOS transistors in the NOT gate schematic.

1.4.2 Transistor-Level Schematic of the CMOS NOT Gate

The PMOS source was connected to the VDD pin and the NMOS source was connected to the VSS pin, while the drains of both transistors were tied together to form the output net (Vout); the gates of both transistors were tied together to form the input net (Vin). The complete transistor-level schematic of the CMOS NOT gate, with all pins and wires labelled, is shown in Figure-1.1. The schematic clearly shows the PMOS pull-up device (PM0, gpdk090_pmos1v, w=720n, l=100n) at the top and the NMOS pull-down device (NM0, gpdk090_nmos1v, w=240n, l=100n) at the bottom, with the shared gate connected to Vin and the shared drain connected to Vout.

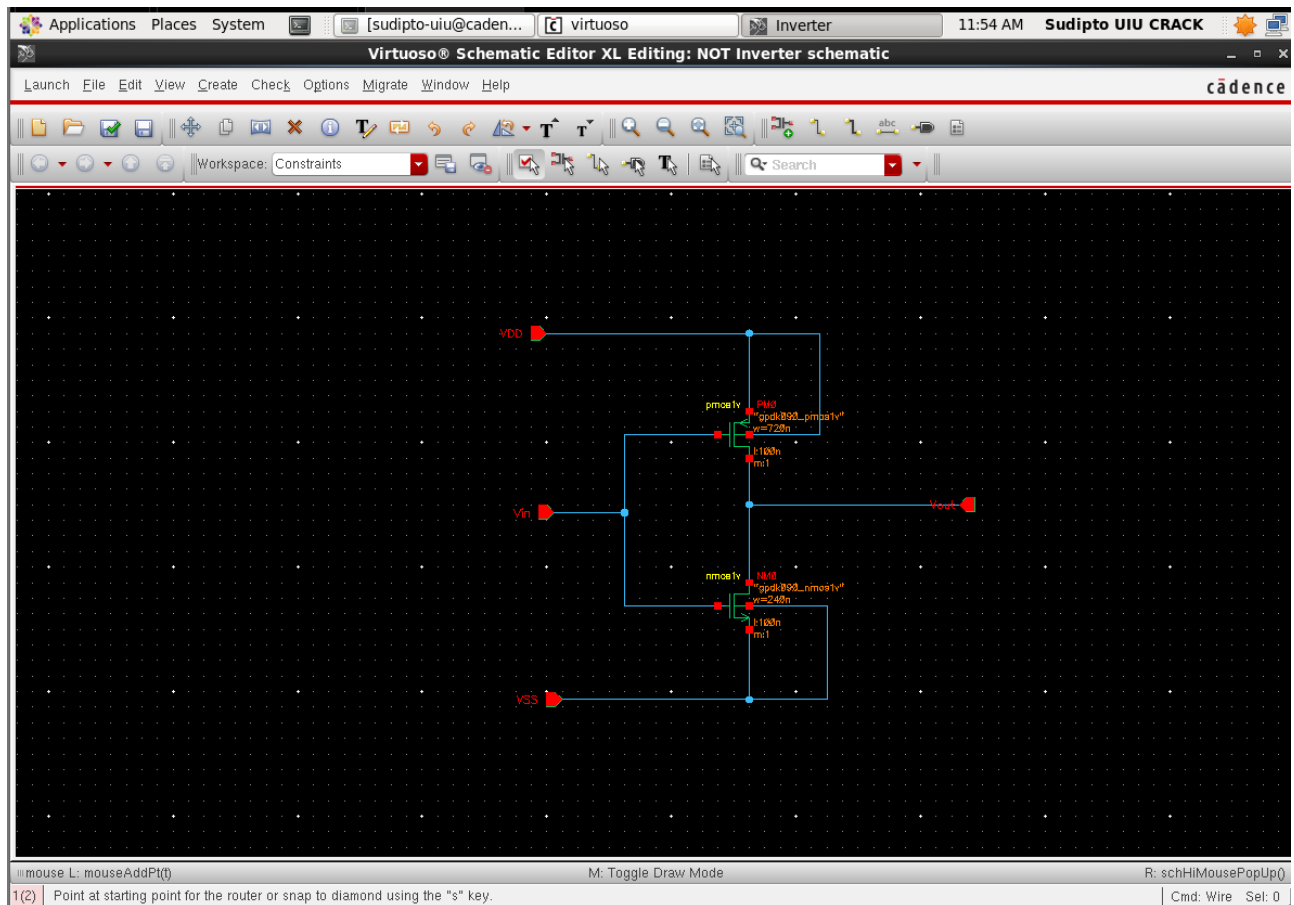


Figure-1.1. Complete transistor-level schematic of the CMOS NOT gate (NOT Inverter), showing the PMOS pull-up network (PM0, $w=720n$), the NMOS pull-down network (NM0, $w=240n$), and the V_{in} , V_{out} , V_{DD} and V_{SS} pins.

1.4.3 Schematic Symbol Generation and Test Bench Construction

After the schematic was checked and saved without errors, a schematic symbol was generated from the cellview using Create → Cellview → From Cellview. The default rectangular shape was replaced with the conventional triangle-and-bubble symbol for an inverter, with V_{in} placed as the left pin, V_{out} as the right pin, and V_{DD} and V_{SS} as the top and bottom pins respectively. A separate schematic, NOT Inverter_test, was then created to verify the NOT gate. The generated inverter symbol (labelled NOT) was instantiated, and a pulse voltage source (vpulse) from analogLib was connected to its input with voltage1 = 0 V, voltage2 = 1.8 V, period = 20 ns and pulse width = 10 ns. A DC voltage source (vdc) set to 1.8 V and a ground (gnd) were connected to the V_{DD} and V_{SS} pins respectively, and the input and output nets were named V_{in} and V_{out} using Create → Wire Name. The completed test bench, with the NOT gate symbol, all sources and net labels, is shown in Figure-1.2.

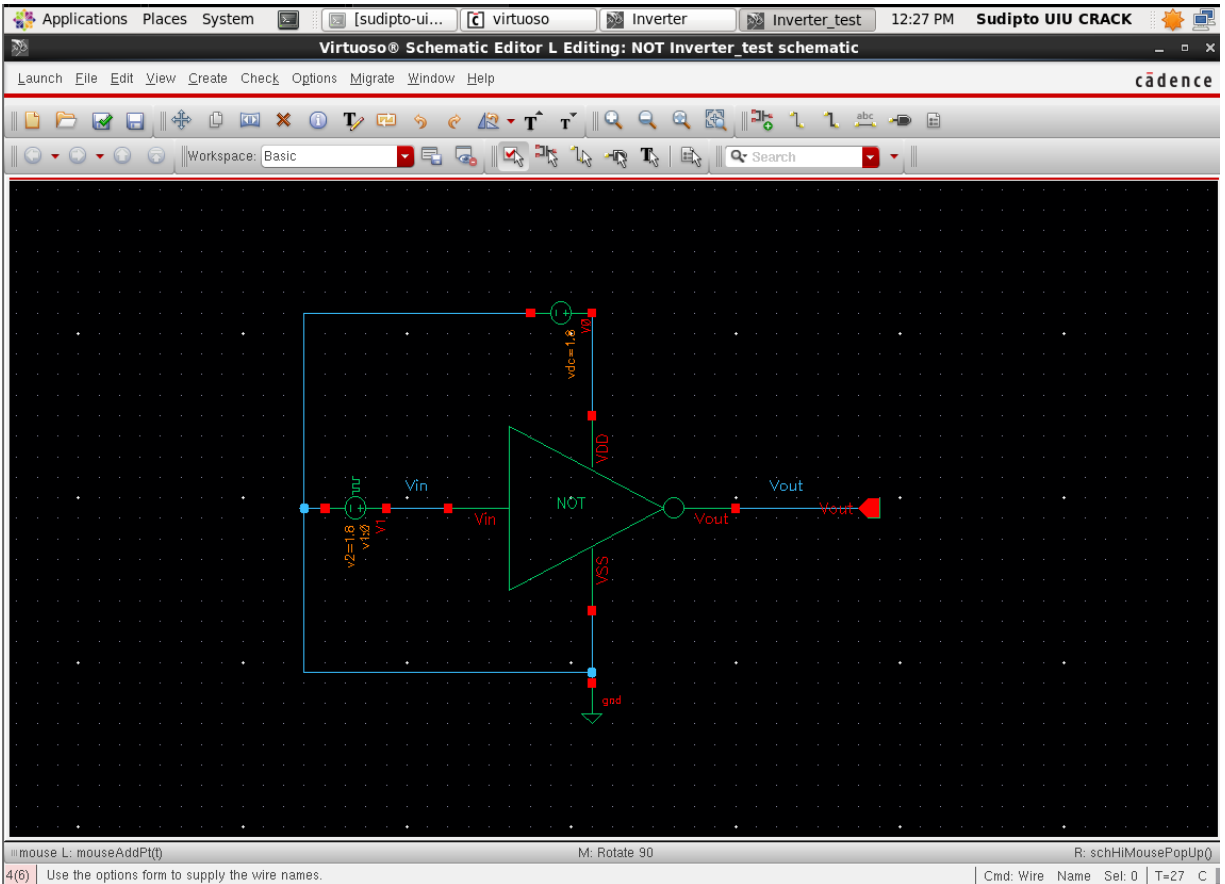


Figure-1.2. Test bench schematic (NOT Inverter_test) showing the instantiated NOT gate symbol (triangle with output bubble), the pulse input source (vpulse, 0–1.8 V, 20 ns period), the DC supply (vdc = 1.8 V), ground, and the Vin and Vout nets labeled.

1.4.4 Simulation Setup

The test bench was simulated in ADE L using the Spectre simulator, with the gpd090 Spectre model library added under Setup → Model Libraries. A transient analysis was configured with a stop time of 100 ns, and a DC analysis was configured to sweep the DC voltage of the vpulse source from 0 V to 2.0 V, with the operating point saved. The input net Vin and the output net Vout were selected as the outputs to be plotted, and the simulation was run using Simulation → Netlist and Run.

1.4.5 Simulation Results: Input and Output Waveforms

Figure-1.3 shows the simulated transient response and the DC voltage transfer characteristic (VTC) of the NOT gate. In the transient response (left panel), Vin is a periodic pulse waveform switching between 0 V and 1.8 V with a 20 ns period and a 10 ns pulse width; Vout is observed to switch in the opposite sense, remaining close to 1.8 V while Vin is LOW and close to 0 V while Vin is HIGH across the full 100 ns simulation window, confirming correct NOT gate operation in every cycle. In the DC response (right panel), Vout decreases monotonically as Vin is swept from 0 V to 2.0 V, forming the characteristic S-shaped VTC curve. The switching threshold voltage VM, the point where the Vin diagonal line intersects the falling Vout curve, is

observed close to the midpoint of the supply range (approximately $V_{DD}/2 \approx 0.9$ V), consistent with the 3:1 PMOS-to-NMOS sizing ratio used in the design.



Figure-1.3. Simulation results: (left) transient response over 100 ns showing the pulse input V_{in} (red) and the inverted output V_{out} (green), and (right) the DC voltage transfer characteristic (VTC) of the CMOS NOT gate with the switching threshold near $V_{DD}/2$.

1.5 Discussion

The discussion on the designed CMOS inverter circuit confirms its simulated behavior aligns with theoretical expectations. Transient simulations validated that the output waveform is the logical complement of the input pulse across a 100 ns window, consistent with the truth table. The DC simulation revealed a distinct voltage transfer characteristic with a sharp transition, and the switching threshold voltage was near the expected $V_{DD}/2$ (approximately 0.9 V), successfully demonstrating that the PMOS transistor, sized three times wider than the NMOS, effectively addressed the PMOS's lower hole mobility. The circuit's design ensures negligible static power dissipation due to the pull-up and pull-down networks never being simultaneously active in steady-state operation.

Minor discrepancies in the switching threshold from the desired midpoint are attributed to device modeling and parasitic effects in the gpd090 technology library. Extending the DC sweep to 2.0 V confirmed the output achieved optimal rail-to-rail levels in both logic states.

1.6 Conclusion

In this experiment, the CMOS NOT gate was designed at the transistor level using Cadence Virtuoso with NMOS and PMOS devices from the gpd090 library, and its schematic symbol generated. The constructed test bench included a pulse input and a 1.8 V DC supply, and 100 ns transient and DC simulations with Spectre in ADE L verified the circuit's performance. Waveform observations confirmed the circuit's effective execution of the logical NOT operation with rail-to-rail outputs. The extracted voltage transfer characteristic validated the transistor sizing strategy, confirming the efficient balance of drive strengths with the 3:1 PMOS-to-NMOS

width ratio. This hands-on experiment enhanced understanding of the standard-cell schematic design flow foundational for future digital and mixed-signal IC designs.

Experiment No-02

Study of CMOS NAND Gate Schematics in Cadence Virtuoso Software

1.1 Objectives

The main objectives of this experiment are summarized as:

- To understand the concept and operation of a two-input CMOS NAND gate as a fundamental building block of digital logic circuits.
- To become familiar with the Cadence Virtuoso schematic editor and the standard IC design flow used to create standard cells.
- To design the transistor-level schematic of a two-input CMOS NAND gate, using two parallel PMOS transistors as the pull-up network and two series NMOS transistors as the pull-down network from the gpd90 technology library, and to generate its schematic symbol.
- To construct a test bench for the NAND gate, with two independent pulse sources at its two inputs, and verify its functionality through transient simulation using Spectre in the Analog Design Environment (ADE L).
- To obtain and analyze the transient input/output wave-forms of the designed NAND gate for all combinations of its two input signals, and to verify them against the NAND truth table.

Keywords- *CMOS NAND gate, NMOS, PMOS, Cadence Virtuoso, schematic editor, Spectre, transient response, truth table, standard cell.*

1.2 Theory

A NAND gate is a fundamental digital logic gate whose output is the complement of the logical AND of its inputs. For a two-input NAND gate with inputs A and B and output Y, the operation is expressed by the Boolean expression $Y = (A \cdot B)'$. The output is LOW (logic 0) only when both inputs are HIGH (logic 1); for every other combination of inputs, the output is HIGH (logic 1). The truth table of an ideal two-input NAND gate is shown in Table-1.1.

A	B	$Y = (A \cdot B)'$
0	0	1
0	1	1
1	0	1
1	1	0

Table-1.1. Truth table of an ideal two-input NAND gate.

1.2.1 The CMOS NAND Gate

A Complementary Metal-Oxide-Semiconductor (CMOS) NAND gate implements the NAND function using two complementary networks of transistors: a pull-up network of two PMOS transistors connected in parallel between the output node and the supply voltage VDD, and a pull-down network of two NMOS transistors connected in series between the output node and ground, VSS. The two inputs, A and B, are each connected to the gate of one PMOS transistor and one series NMOS transistor, and the drains of the top PMOS device and the top NMOS device are tied together to form the single output terminal.

The output node is pulled down to ground only when both series NMOS transistors are simultaneously ON, which occurs only when both A and B are HIGH; for every other input combination, at least one of the two parallel PMOS transistors is ON, pulling the output up to VDD. Because the pull-up and pull-down networks are never both fully ON at the same time for any static input combination, there is no direct conduction path from VDD to ground, and the static power dissipation of the CMOS NAND gate is ideally zero; power is dissipated mainly during switching (dynamic power) and due to leakage currents in real devices.

1.2.2 Transistor Sizing and the Effect of NMOS Stacking

Unlike a single-input inverter, the pull-down network of a two-input NAND gate consists of two NMOS transistors connected in series (a transistor “stack”). Because the two series devices carry the same current, the effective ON-resistance of the pull-down network is higher than that of a single NMOS transistor of the same width, which tends to slow the falling (HIGH-to-LOW) transition of the output relative to the rising transition.

Because the mobility of holes in a PMOS transistor is lower than the mobility of electrons in an NMOS transistor, a PMOS device must be made wider than an NMOS device of the same length to provide comparable drive strength. In this experiment, the same transistor dimensions used for the earlier CMOS inverter design were retained for the NAND gate: each PMOS device was sized with a total width of 720 nm, and each NMOS device with a total width of 240 nm, for the same channel length of 100 nm, giving the same 3:1 PMOS-to-NMOS width ratio as before. Keeping these dimensions unchanged preserves consistency with the earlier experiment, although the series NMOS stack means the pull-down network is comparatively weaker here than it was for the single-transistor inverter.

1.2.3 Cadence Virtuoso Design Flow

In this experiment, the CMOS NAND gate is designed and verified using the Cadence Virtuoso IC design environment. The overall flow consists of creating a design library attached to a process technology library (gpd090), drawing the transistor-level schematic in the Schematic XL editor by placing and sizing two PMOS and two NMOS instances and wiring them into the parallel pull-up / series pull-down NAND topology, generating a schematic symbol for the NAND cell, and finally building a separate test bench schematic in which the NAND symbol is instantiated along with two independent input sources and supply sources. The test bench is then simulated using the Spectre simulator inside the Analog Design Environment (ADE L) to verify the NAND gate's transient behaviour.

1.3 Hardware/Software Used for the Experiment

1. Cadence Virtuoso Schematic Editor (Schematic XL)
2. Virtuoso Analog Design Environment (ADE L)
3. Spectre circuit simulator
4. gpd090 generic process design kit (technology library)
5. Two NMOS and two PMOS transistors (gpd090_nmos1v, gpd090_pmos1v)
6. Two pulse voltage sources (vpulse), a DC voltage source (vdc) and ground (gnd) from analogLib
7. Linux workstation running the Cadence design environment

1.4 Experimental and Simulation Analysis

A design library (NAND_Gate) was created and attached to the gpd090 technology library, and a schematic cellview (carried over from the earlier inverter template and therefore still internally named inverter) was modified within it to implement the two-input NAND function. Two NMOS and two PMOS instances were placed in the schematic, sized as described in Section 1.2.2, and wired into the parallel PMOS pull-up network and series NMOS pull-down network, with two input pins (A and B), an output pin, and the VDD/VSS supply pins. A schematic symbol was then generated for the NAND cell. A separate test schematic (inverter_test) was built in which the NAND symbol was instantiated with two independent pulse sources at its two inputs and a DC supply and ground connected to its power pins, and the design was simulated in ADE L using the Spectre simulator. Each of these steps is illustrated below.

1.4.1 Transistor Sizing and Parameter Selection

The width and length of each transistor were set in the Add Instance window at the time of placement. As in the earlier inverter design, each NMOS transistor (NM0, NM1) was sized with a total width of 240 nm and a channel length of 100 nm, while each PMOS transistor (PM0, PM1) was sized with a total width of 720 nm and the same channel length of 100 nm, giving the 3:1 PMOS-to-NMOS width ratio discussed in Section 1.2.2. The parameter selection window used for setting the NMOS device dimensions is shown in Figure-1.1.



Figure-1.1. Add Instance window showing the width and length parameters set for the NMOS transistor (Total Width = 240n, Length = 100n).

Device	Width (W)	Length (L)
NMOS – NM0, NM1 (gpd090_nmos1v)	240 nm	100 nm
PMOS – PM0, PM1 (gpd090_pmos1v)	720 nm	100 nm

Table-1.2. Device parameters used for the NMOS and PMOS transistors in the NAND gate schematic.

1.4.2 Transistor-Level Schematic of the CMOS NAND Gate

The two PMOS transistor sources were connected to the VDD pin and connected in parallel between VDD and the output node, while the two NMOS transistors were connected in series between the output node and the VSS pin, with the drains of the top PMOS and top NMOS devices tied together to form the output net (Vout). The gates of one PMOS and one series NMOS were tied together to form input A (Vin1), and the gates of the other PMOS and NMOS were tied together to form input B (Vin2). The complete transistor-level schematic of the CMOS NAND gate, with all pins and wires labelled, is shown in Figure-1.2.

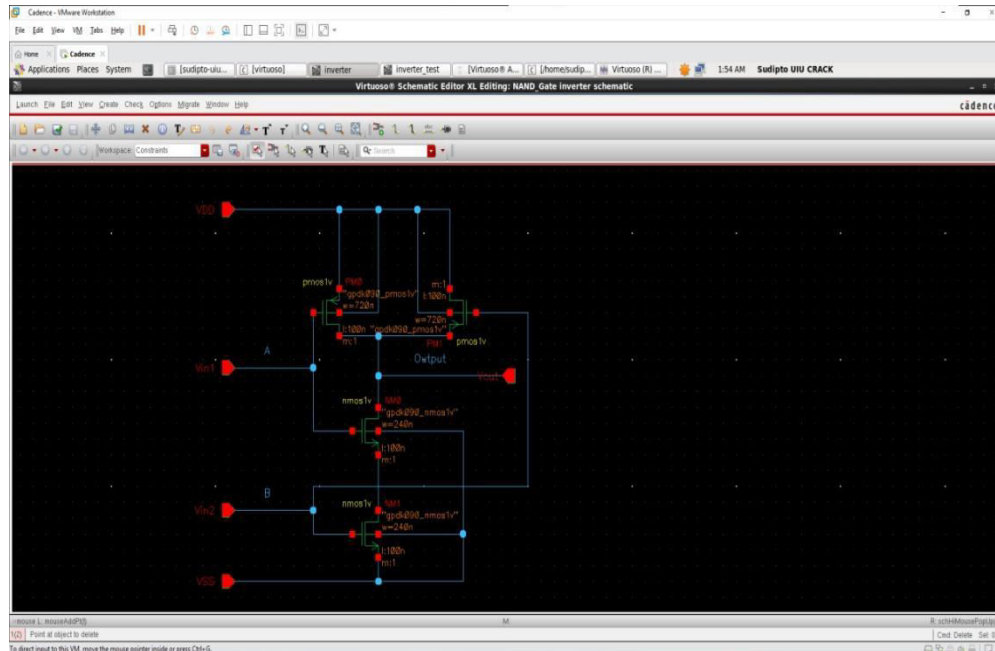


Figure-1.2. Complete transistor-level schematic of the CMOS NAND gate, showing the two parallel PMOS transistors (PM0, PM1) forming the pull-up network, the two series NMOS transistors (NM0, NM1) forming the pull-down network, and the A, B, Vout, VDD and VSS pins.

1.4.3 Schematic Symbol Generation

After the schematic was checked and saved without errors, a schematic symbol was generated from the cellview using Create → Cellview → From Cellview, with Vin1 (A) and Vin2 (B) placed as the two left-side input pins, Vout placed as the right-side output pin, and VDD and VSS placed as the top and bottom pins. The default shape generated by Virtuoso was replaced with the conventional two-input NAND symbol, a curved AND-gate body with an output bubble, as shown in Figure-1.3.

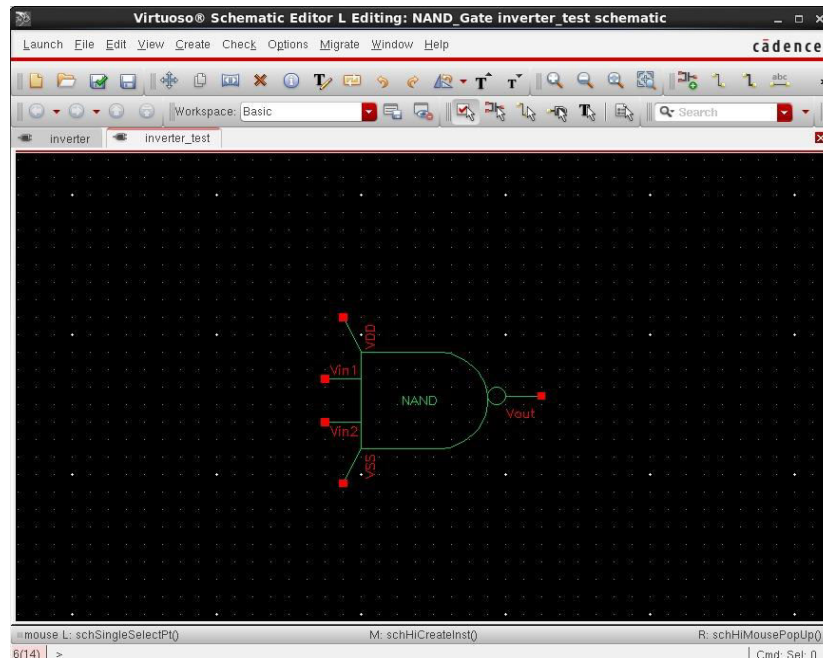


Figure-1.3. Generated schematic symbol of the CMOS NAND gate, showing the two inputs Vin1 and Vin2, the output Vout, and the VDD and VSS supply pins.

1.4.4 Test Bench Construction and Source Addition

A separate schematic, *inverter_test*, was created to verify the NAND gate. The generated NAND symbol was instantiated, and two independent pulse voltage sources (*vpulse*) from *analogLib* were connected to its two inputs: source V1 was connected to input A (Vin1) with voltage1 = 0 V, voltage2 = 1.8 V, period = 20 ns and pulse width = 10 ns, while source V2 was connected to input B (Vin2) with voltage1 = 0 V, voltage2 = 1.8 V, period = 40 ns and pulse width = 20 ns, so that all four input combinations of A and B are exercised over the course of the simulation. A DC voltage source (*vdc*) set to 1.8 V and a ground (*gnd*) were connected to the VDD and VSS pins respectively, and the input and output nets were named Vin1, Vin2 and Vout using Create → Wire Name. The completed test bench, with all sources added, is shown in Figure-1.4.

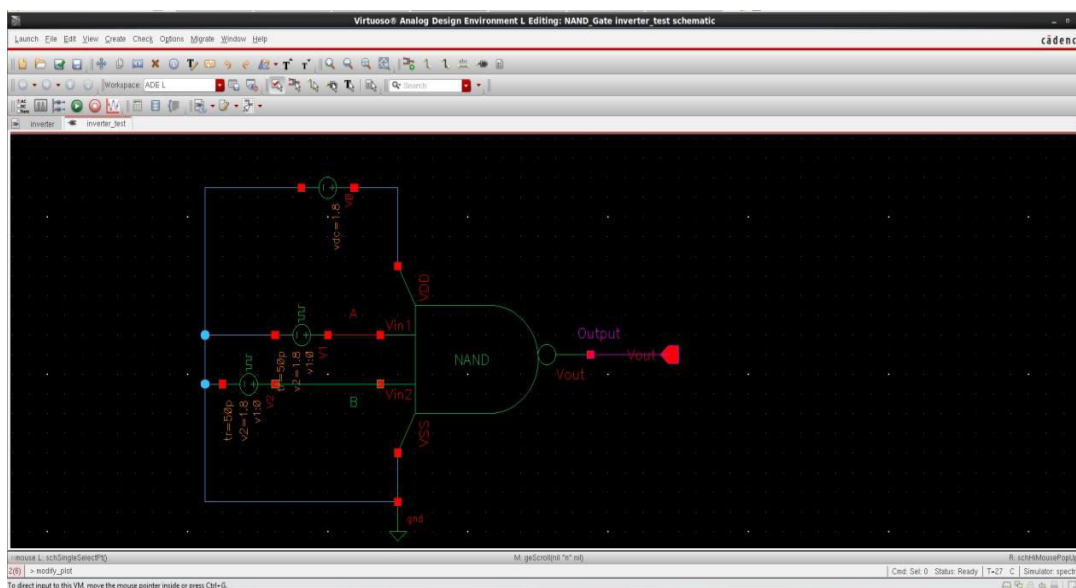


Figure-1.4. Test bench schematic (inverter_test) with the two pulse input sources V1 and V2, the DC supply, ground, and the NAND symbol, with the Vin1, Vin2 and Vout nets labelled.

1.4.5 Input Source Parameter Selection

The Add Instance windows used to configure the two pulse sources and the DC supply are shown in Figure-1.5, Figure-1.6 and Figure-1.7. Figure-1.5 shows the parameters of source V1 driving input A, with Voltage 1 = 0 V, Voltage 2 = 1.8 V, a period of 20 ns, a pulse width of 10 ns, and rise/fall times of 50 ps. Figure-1.6 shows the parameters of source V2 driving input B, configured identically except for a longer period of 40 ns and a pulse width of 20 ns, so that B toggles at half the frequency of A. Figure-1.7 shows the Add Instance window for the vdc source used to set the VDD supply to a DC voltage of 1.8 V.

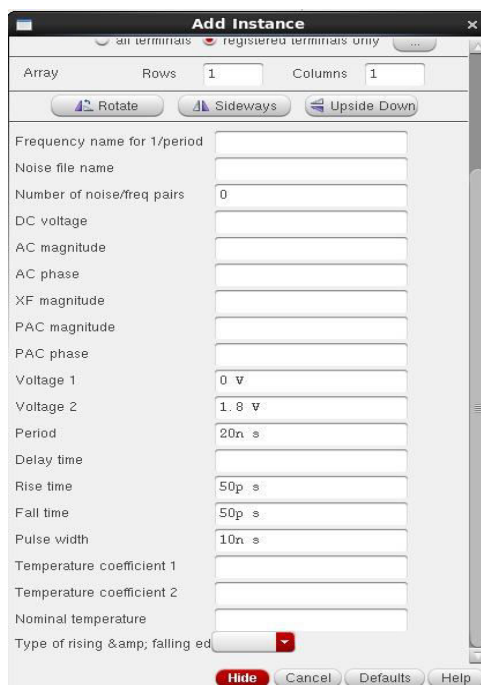


Figure-1.5. Add Instance window showing the pulse source V1 (NAND input A): Voltage 1 = 0 V, Voltage 2 = 1.8 V, period = 20 ns, pulse width = 10 ns.

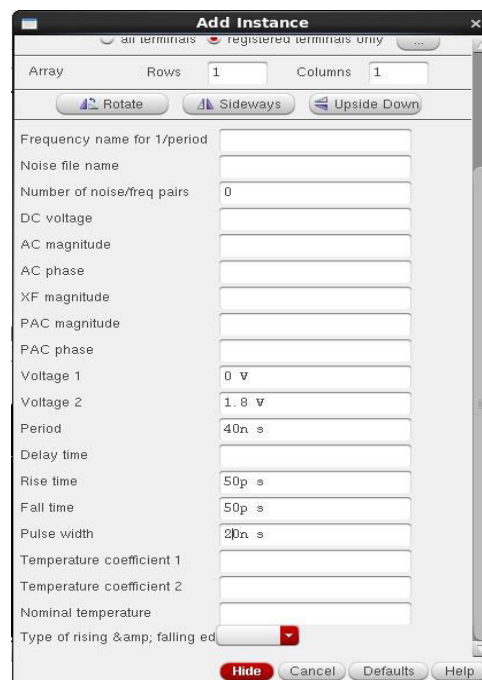


Figure-1.6. Add Instance window showing the pulse source V2 (NAND input B): Voltage 1 = 0 V, Voltage 2 = 1.8 V, period = 40 ns, pulse width = 20 ns.

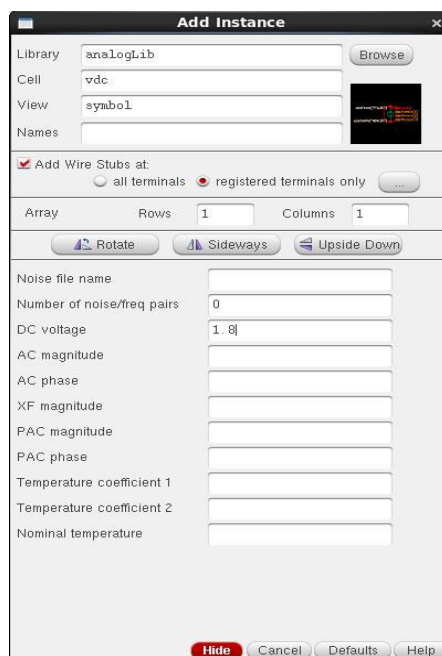


Figure-1.7. Add Instance window showing the vdc source used for the VDD supply, set to a DC voltage of 1.8 V.

1.4.6 Simulation Setup

The test bench was simulated in ADE L using the Spectre simulator, with the gpd090 Spectre model library added under Setup → Model Libraries. A transient analysis was configured with a stop time of 200 ns so that several cycles of both the 20 ns-period input A and the 40 ns-period input B were captured. The input nets A and B and the output net Vout were selected as the outputs to be plotted, and the simulation was run using Simulation → Netlist and Run.

1.4.7 Simulation Results: Input and Output Wave-forms

Figure-1.8 shows the simulated transient response of the NAND gate. In this response, A is a periodic pulse waveform switching between 0 V and 1.8 V with a 20 ns period and a 10 ns pulse width, and B is a periodic pulse waveform switching between 0 V and 1.8 V with a 40 ns period and a 20 ns pulse width, so that the two inputs together step through all four logic combinations — (0,0), (0,1), (1,0) and (1,1) — over the course of the simulation. The output Vout is observed to remain HIGH for every input combination except when A and B are simultaneously HIGH, during which Vout switches LOW, in direct agreement with the two-input NAND truth table given in Table-1.1.

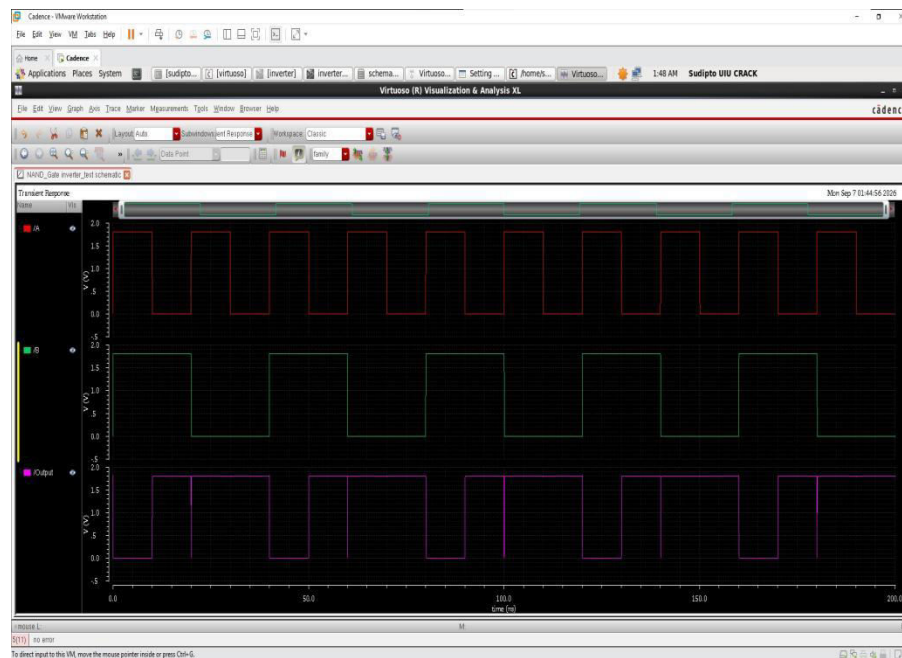


Figure-1.8. Simulation results: transient response showing the two pulse inputs A and B and the resulting NAND output Vout.

1.5 Discussion

The simulated behaviour of the designed circuit was found to be in agreement with the theoretical operation of a CMOS NAND gate described in Section 1.2. The transient simulation confirmed that the output was pulled LOW only during the intervals when input A and input B were simultaneously HIGH, and remained HIGH for the other three input combinations, matching the expected truth table (Table-1.1) in every cycle. Because the series NMOS pull-down network and the parallel PMOS pull-up network are never both fully ON for the same static input combination, the circuit is also expected to dissipate negligible static power, consistent with the fundamental advantage of CMOS logic. Retaining the same transistor dimensions used for the earlier CMOS

inverter ($W_P = 720$ nm, $W_N = 240$ nm, $L = 100$ nm for all devices) kept the design consistent with that experiment, although the NMOS stack in the pull-down path is expected to make the HIGH-to-LOW transition somewhat slower than in the single-transistor inverter case; this trade-off did not affect the correctness of the logical output observed in simulation.

1.6 Conclusion

In this experiment, a two-input CMOS NAND gate was successfully designed at the transistor level in Cadence Virtuoso, using two parallel PMOS transistors as the pull-up network and two series NMOS transistors as the pull-down network from the gpdk090 technology library, and its schematic symbol was generated. A test bench was constructed with two independent pulse sources at the two inputs and a DC supply, and the circuit was verified through transient simulation using Spectre in ADE L. The observed input-output wave-forms confirmed that the circuit correctly performs the logical NAND operation, producing a LOW output only when both inputs were simultaneously HIGH, consistent with the NAND truth table. This experiment provided practical, hands-on experience with extending a basic standard-cell schematic design flow from a single-input inverter to a multi-input logic gate, which forms the basis for more complex digital and mixed-signal IC design in subsequent experiments.

Experiment No-03

Study of CMOS AND Gate Schematics in Cadence Virtuoso Software

1.1 Objectives

The main objectives of this experiment are summarized as:

- To understand the concept and operation of a two-input CMOS AND gate as a fundamental building block of digital logic circuits.
- To become familiar with the Cadence Virtuoso schematic editor and the standard IC design flow used to create standard cells.
- To design the transistor-level schematic of a two-input CMOS AND gate, implemented as a NAND stage (two parallel PMOS transistors as the pull-up network and two series NMOS transistors as the pull-down network) followed by a CMOS inverter stage, using devices from the gpdk090 technology library, and to generate its schematic symbol.
- To construct a test bench for the AND gate, with two independent pulse sources at its two inputs, and verify its functionality through transient simulation using Spectre in the Analog Design Environment (ADE L).
- To obtain and analyze the transient input/output wave-forms of the designed AND gate for all combinations of its two input signals, and to verify them against the AND truth table.

Keywords- CMOS AND gate, NMOS, PMOS, Cadence Virtuoso, schematic editor, Spectre, NAND-inverter cascade, transient response, truth table, standard cell.

1.2 Theory

An AND gate is a fundamental digital logic gate whose output is the logical AND of its inputs. For a two-input AND gate with inputs A and B and output Y, the operation is expressed by the Boolean expression $Y = A \cdot B$. The output is HIGH (logic 1) only when both inputs are HIGH (logic 1); for every other combination of inputs, the output is LOW (logic 0). The truth table of an ideal two-input AND gate is shown in Table-1.1.

A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

Table-1.1. Truth table of an ideal two-input AND gate.

1.2.1 The CMOS AND Gate (NAND-Inverter Cascade)

Unlike the inverter and the NAND gate, a single stage of complementary CMOS transistors cannot directly realize a non-inverting AND function, because a purely complementary pull-up/pull-down network always produces an inverted (NAND-type) output. A two-input CMOS AND gate is therefore built as a cascade of two stages: a first stage that is a standard two-input NAND gate, consisting of two PMOS transistors (PM0, PM1) connected in parallel between VDD and an internal node, and two NMOS transistors (NM0, NM1) connected in series between that internal node and VSS; and a second stage that is a CMOS inverter, consisting of one PMOS transistor

(PM2) and one NMOS transistor (NM2), which inverts the internal NAND node to produce the final output Vout.

The internal node between the two stages is pulled LOW only when both series NMOS transistors of the first stage are simultaneously ON, which happens only when both A and B are HIGH; for every other input combination, the internal node is pulled HIGH by the parallel PMOS pull-up network. The inverter stage then inverts this internal node, so the final output Vout is HIGH only when both A and B are HIGH, and LOW for every other input combination the correct AND function. Because neither the NAND stage nor the inverter stage ever has its pull-up and pull-down networks both fully ON for the same static input combination, the static power dissipation of the overall CMOS AND gate is ideally zero, just as for the individual NAND and inverter stages.

1.2.2 Transistor Sizing and the Two-Stage Cascade

As in the standalone NAND gate, the pull-down network of the first (NAND) stage consists of two series NMOS transistors (a transistor “stack”), which gives the internal node a higher effective pull-down resistance than a single NMOS transistor of the same width would provide, tending to slow its HIGH-to-LOW transition. The second (inverter) stage, however, uses only a single PMOS and a single NMOS transistor, so it switches with the same speed characteristics as a standalone CMOS inverter and restores a full-swing, sharply-transitioning output at Vout.

Because the mobility of holes in a PMOS transistor is lower than the mobility of electrons in an NMOS transistor, a PMOS device must be made wider than an NMOS device of the same length to provide comparable drive strength. In this experiment, the same transistor dimensions used for the earlier CMOS inverter and NAND gate designs were retained for all three PMOS and all three NMOS devices of the AND gate: every PMOS device (PM0, PM1, PM2) was sized with a total width of 720 nm, and every NMOS device (NM0, NM1, NM2) with a total width of 240 nm, for the same channel length of 100 nm, giving the same 3:1 PMOS-to-NMOS width ratio as before. Keeping these dimensions unchanged preserves consistency with the earlier experiments for both the NAND stage and the inverter stage.

1.2.3 Cadence Virtuoso Design Flow

In this experiment, the CMOS AND gate is designed and verified using the Cadence Virtuoso IC design environment. The overall flow consists of creating a design library attached to a process technology library (gpd090), drawing the transistor-level schematic in the Schematic XL editor by placing and sizing three PMOS and three NMOS instances and wiring them into a NAND stage followed by an inverter stage, generating a schematic symbol for the AND cell, and finally building a separate test bench schematic in which the AND symbol is instantiated along with two independent input sources and supply sources. The test bench is then simulated using the Spectre simulator inside the Analog Design Environment (ADE L) to verify the AND gate’s transient behaviour.

1.3 Hardware/Software Used for the Experiment

1. Cadence Virtuoso Schematic Editor (Schematic XL)
2. Virtuoso Analog Design Environment (ADE L)
3. Spectre circuit simulator
4. gpd090 generic process design kit (technology library)
5. Three NMOS and three PMOS transistors (gpd090_nmos1v, gpd090_pmos1v)
6. Two pulse voltage sources (vpulse), a DC voltage source (vdc) and ground (gnd) from analogLib
7. Linux workstation running the Cadence design environment

1.4 Experimental and Simulation Analysis

A design library (AND_Gate) was created and attached to the gpd090 technology library, and a schematic cellview (carried over from the earlier inverter/NAND template and therefore still internally named inverter) was modified within it to implement the two-input AND function. Three NMOS and three PMOS instances were placed in the schematic, sized as described in Section 1.2.2, and wired into a first NAND stage (PM0, PM1, NM0, NM1) followed by a second inverter stage (PM2, NM2), with two input pins (A and B), an output pin, and the VDD/VSS supply pins. A schematic symbol was then generated for the AND cell. A separate test schematic (inverter_test) was built in which the AND symbol was instantiated with two independent pulse sources at its two inputs and a DC supply and ground connected to its power pins, and the design was simulated in ADE L using the Spectre simulator. Each of these steps is illustrated below.

1.4.1 Transistor Sizing and Parameter Selection

The width and length of each transistor were set in the Add Instance window at the time of placement. As in the earlier inverter and NAND designs, each NMOS transistor (NM0, NM1, NM2) was sized with a total width of 240 nm and a channel length of 100 nm, while each PMOS transistor (PM0, PM1, PM2) was sized with a total width of 720 nm and the same channel length of 100 nm, giving the 3:1 PMOS-to-NMOS width ratio discussed in Section 1.2.2. The parameter selection window used for setting the NMOS device dimensions is shown in Figure-1.1.



Figure-1.1. Add Instance window showing the width and length parameters set for the NMOS transistor (Total Width = 240n, Length = 100n).

Device	Width (W)	Length (L)
NMOS – NM0, NM1, NM2 (gpd090_nmos1v)	240 nm	100 nm
PMOS – PM0, PM1, PM2 (gpd090_pmos1v)	720 nm	100 nm

Table-1.2. Device parameters used for the NMOS and PMOS transistors in the AND gate schematic.

1.4.2 Transistor-Level Schematic of the CMOS AND Gate

The two PMOS transistors of the first stage (PM0, PM1) were connected to the VDD pin and connected in parallel between VDD and an internal node, while the two NMOS transistors (NM0, NM1) were connected in series between that internal node and the VSS pin. The gates of one PMOS and one series NMOS were tied together to form input A (Vin1), and the gates of the other PMOS and NMOS were tied together to form input B (Vin2). The internal node was then connected to the gates of the second-stage inverter transistors (PM2, NM2), whose drains were tied together to form the output net (Vout), with PM2 connected to VDD and NM2 connected to VSS. The complete transistor-level schematic of the CMOS AND gate, with all pins and wires labeled, is shown in Figure-1.2.

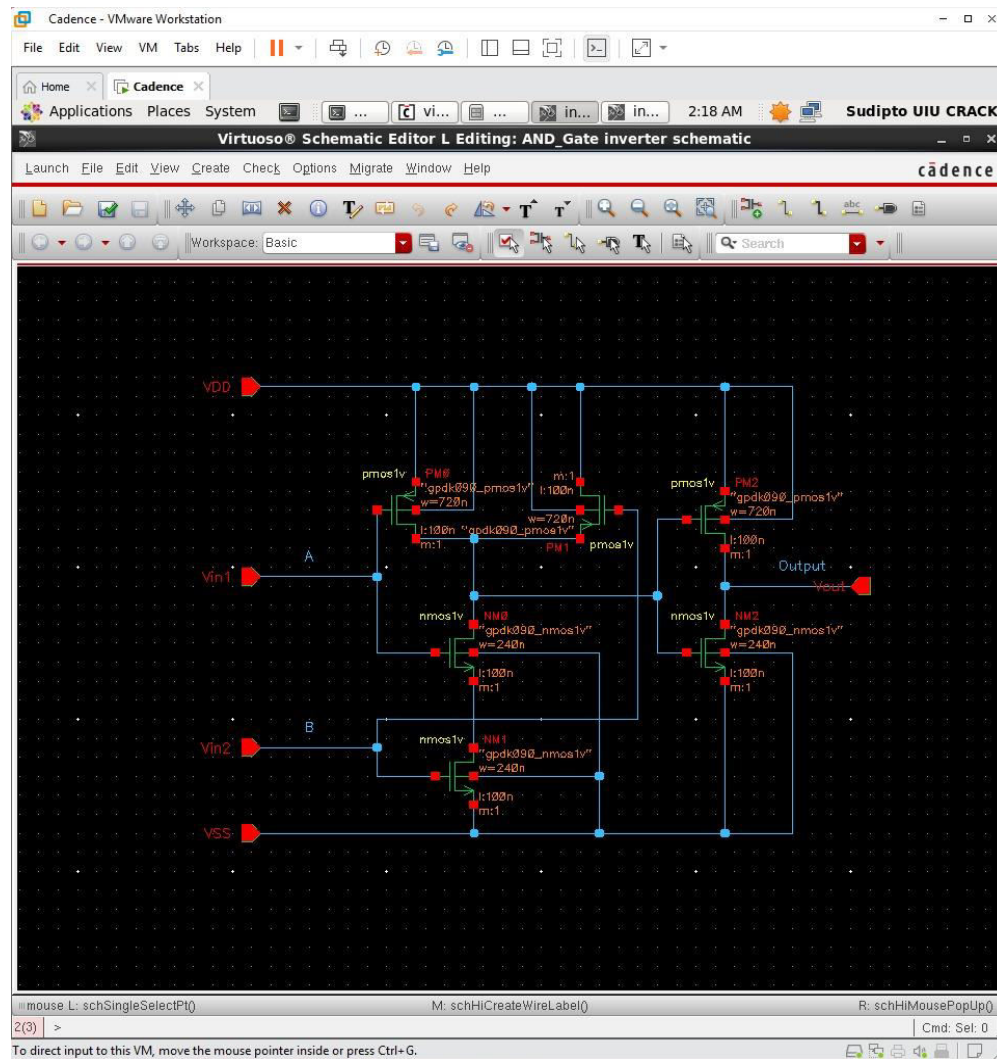


Figure-1.2. Complete transistor-level schematic of the CMOS AND gate, showing the first-stage NAND network (PM0, PM1 in parallel; NM0, NM1 in series) producing an internal node, and the second-stage inverter (PM2, NM2) that inverts this node to produce the final output Vout, with the A, B, Vout, VDD and VSS pins labeled.

1.4.3 Schematic Symbol Generation

After the schematic was checked and saved without errors, a schematic symbol was generated from the cell-view using Create → Cellview → From Cellview, with Vin1 (A) and Vin2 (B) placed as the two left-side input pins, Vout placed as the right-side output pin, and VDD and VSS placed as the top and bottom pins. The default shape generated by Virtuoso was replaced with the conventional two-input AND symbol, a curved AND-gate body without an output bubble (distinguishing it from the NAND symbol used previously), as shown in Figure-1.3.

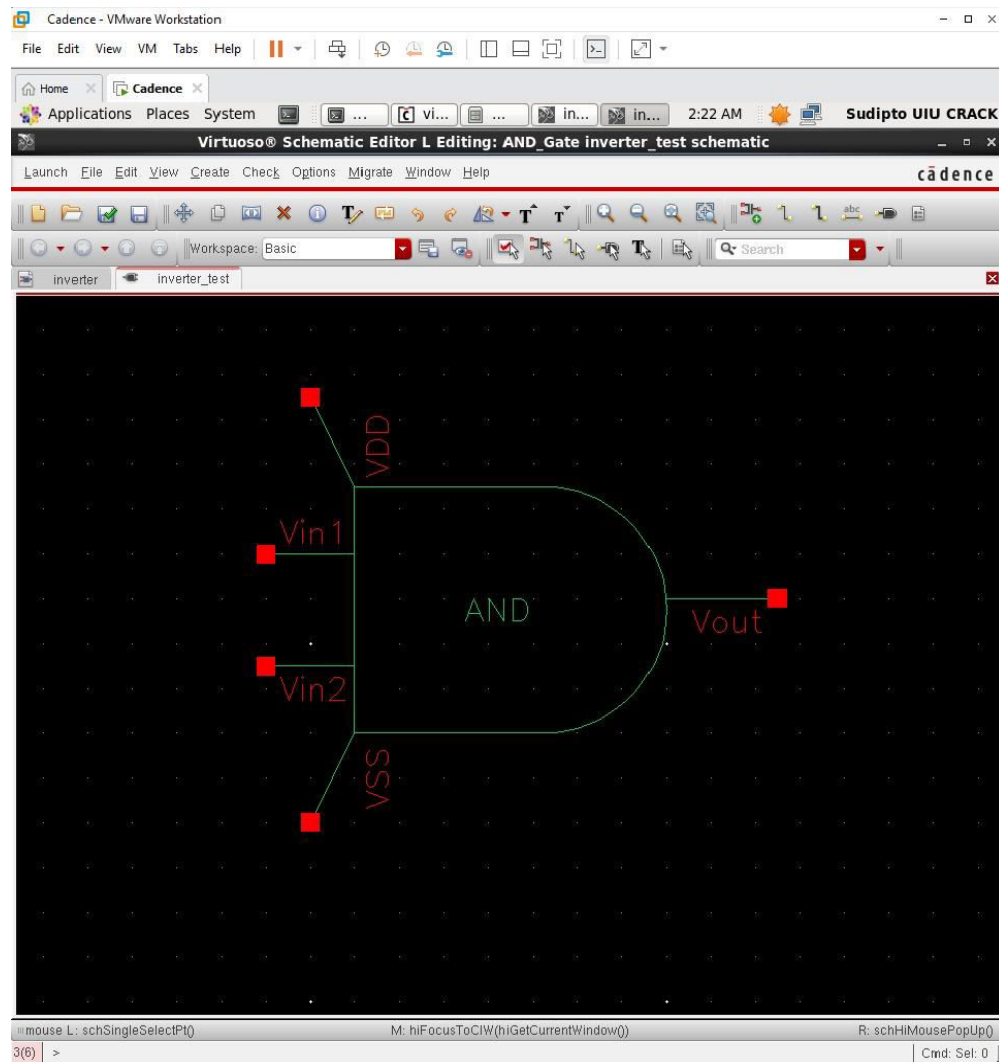


Figure-1.3. Generated schematic symbol of the CMOS AND gate, showing the two inputs Vin1 and Vin2, the output Vout, and the VDD and VSS supply pins.

1.4.4 Test Bench Construction and Source Addition

A separate schematic, `inverter_test`, was created to verify the AND gate. The generated AND symbol was instantiated, and two independent pulse voltage sources (`vpulse`) from `analogLib` were connected to its two inputs: source V1 was connected to input A (Vin1) with $\text{voltage1} = 0 \text{ V}$, $\text{voltage2} = 1.8 \text{ V}$, $\text{period} = 20 \text{ ns}$ and $\text{pulse width} = 10 \text{ ns}$, while source V2 was connected to input B (Vin2) with $\text{voltage1} = 0 \text{ V}$, $\text{voltage2} = 1.8 \text{ V}$, $\text{period} = 40 \text{ ns}$ and $\text{pulse width} = 20 \text{ ns}$, so that all four input combinations of A and B are exercised over the course of the simulation. A DC voltage source (`vdc`) set to 1.8 V and a ground (`gnd`) were connected to the VDD and VSS pins respectively, and the input and output nets were named Vin1, Vin2 and Vout using `Create → Wire Name`. The completed test bench, with all sources added, is shown in Figure-1.4.

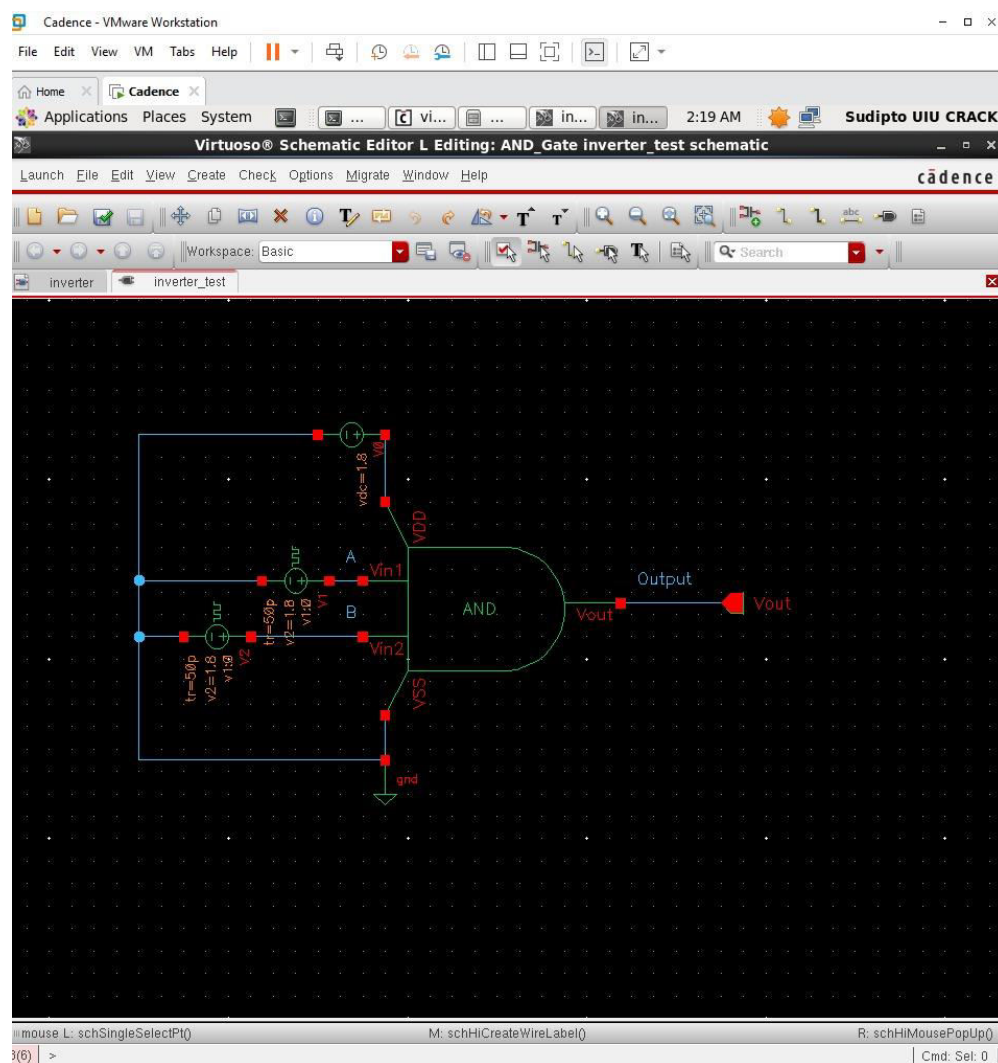


Figure-1.4. Test bench schematic (inverter_test) with the two pulse input sources V1 and V2, the DC supply, ground, and the AND symbol, with the Vin1, Vin2 and Vout nets labeled.

1.4.5 Input Source Parameter Selection

The Add Instance windows used to configure the two pulse sources and the DC supply are shown in Figure-1.5, Figure-1.6 and Figure-1.7. Figure-1.5 shows the parameters of source V1 driving input A, with Voltage 1 = 0 V, Voltage 2 = 1.8 V, a period of 20 ns, a pulse width of 10 ns, and rise/fall times of 50 ps. Figure-1.6 shows the parameters of source V2 driving input B, configured identically except for a longer period of 40 ns and a pulse width of 20 ns, so that B toggles at half the frequency of A. Figure-1.7 shows the Add Instance window for the vdc source used to set the VDD supply to a DC voltage of 1.8 V.

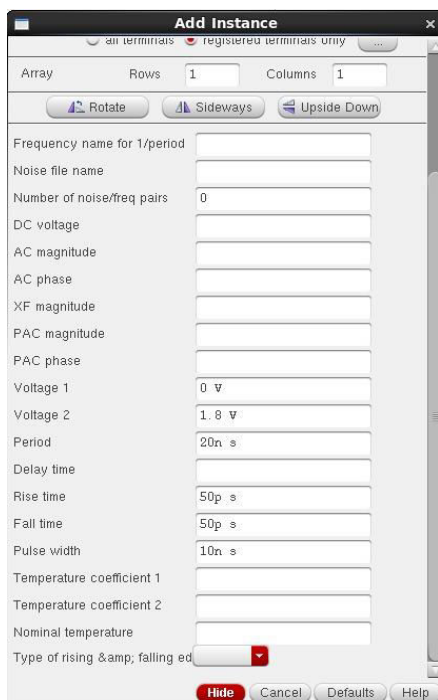


Figure-1.5. Add Instance window showing the pulse source V1 (AND gate input A): Voltage 1 = 0 V, Voltage 2 = 1.8 V, period = 20 ns, pulse width = 10 ns.

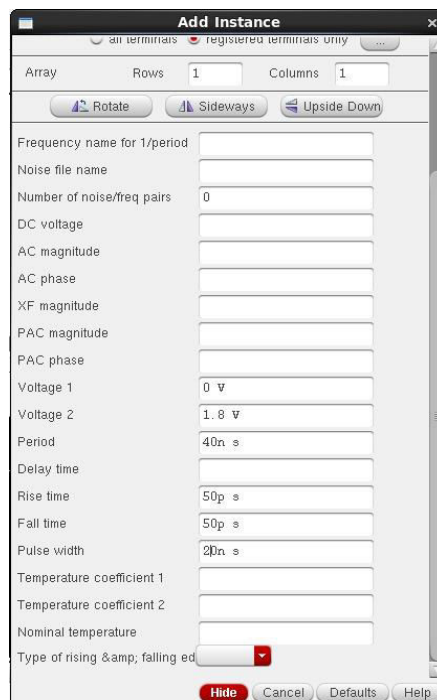


Figure-1.6. Add Instance window showing the pulse source V2 (AND gate input B): Voltage 1 = 0 V, Voltage 2 = 1.8 V, period = 40 ns, pulse width = 20 ns.

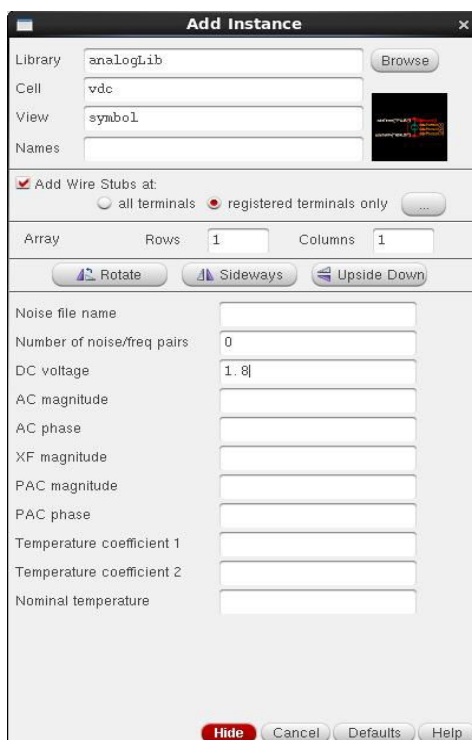


Figure-1.7. Add Instance window showing the vdc source used for the VDD supply, set to a DC voltage of 1.8 V.

1.4.6 Simulation Setup

The test bench was simulated in ADE L using the Spectre simulator, with the gpd090 Spectre model library added under Setup → Model Libraries. A transient analysis was configured with a stop time of 200 ns so that several cycles of both the 20 ns-period input A and the 40 ns-period input B were captured. The input nets A and B and the output net Vout were selected as the outputs to be plotted, and the simulation was run using Simulation → Netlist and Run.

1.4.7 Simulation Results: Input and Output Waveforms

Figure-1.8 shows the simulated transient response of the AND gate. The left-hand plot shows the two pulse inputs, A (top) and B (bottom), each switching between 0 V and 1.8 V, with A having a 20 ns period and a 10 ns pulse width and B having a 40 ns period and a 20 ns pulse width, over an 80 ns window. The right-hand plot shows the corresponding output Vout over a shorter, zoomed-in 30 ns window. Consistent with the AND function, Vout is observed to go HIGH only during the interval where A and B are simultaneously HIGH near the start of the simulation, and to remain LOW for the rest of the observed window whenever A and B are not both HIGH at the same time, in direct agreement with the two-input AND truth table given in Table-1.1.

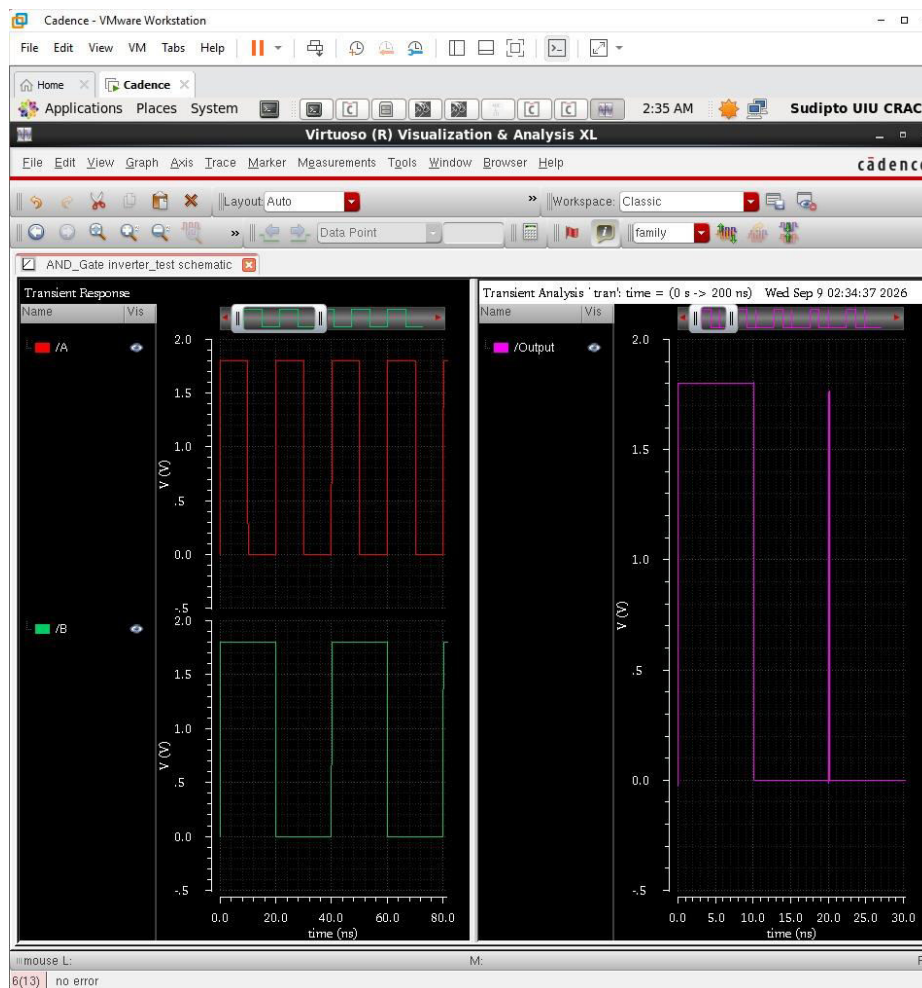


Figure-1.8. Simulation results: the two pulse inputs A and B (left) and the corresponding AND output Vout, shown on a zoomed time axis (right).

1.5 Discussion

The simulated behaviour of the designed circuit was found to be in agreement with the theoretical operation of a CMOS AND gate described in Section 1.2. The transient simulation confirmed that the output V_{out} was HIGH only during the interval where both input A and input B were simultaneously HIGH, and LOW otherwise, matching the expected truth table (Table-1.1). This confirms that cascading a two-input NAND stage with a CMOS inverter stage correctly restores the non-inverted AND function at the output. Because neither stage ever has its pull-up and pull-down networks both fully ON for the same static input combination, the circuit is also expected to dissipate negligible static power, consistent with the fundamental advantage of CMOS logic. Retaining the same transistor dimensions used for the earlier CMOS inverter and NAND gate designs ($W_P = 720$ nm, $W_N = 240$ nm, $L = 100$ nm for all six devices) kept the design consistent with those experiments, although the series NMOS stack in the first stage is expected to make its internal HIGH-to-LOW transition somewhat slower than a single-transistor stage; the added inverter stage did not, however, affect the correctness of the final logical output observed in simulation.

1.6 Conclusion

In this experiment, a two-input CMOS AND gate was successfully designed at the transistor level in Cadence Virtuoso, implemented as a two-input NAND stage (two parallel PMOS and two series NMOS transistors) cascaded with a CMOS inverter stage (one PMOS and one NMOS transistor), using devices from the gpd090 technology library, and its schematic symbol was generated. A test bench was constructed with two independent pulse sources at the two inputs and a DC supply, and the circuit was verified through transient simulation using Spectre in ADE L. The observed input-output waveforms confirmed that the circuit correctly performs the logical AND operation, producing a HIGH output only when both inputs were simultaneously HIGH, consistent with the AND truth table. This experiment provided practical, hands-on experience with building a composite standard cell from two cascaded logic stages (NAND + inverter), which forms the basis for more complex digital and mixed-signal IC design in subsequent experiments.